

Online Assignment Submission

UNIVERSITY OF
BIRMINGHAM

School of Government

Student ID Number: 3059714

Programme: MSc AI and Sustainable Development (2025/2026)

Module: 40109 - AI & Sustainable Development

Assignment Title: AI & Sustainable Development - Assignment 1 - 1000 words worth 75% of overall mark

Date and Time of Submission: 20:15 06/12/2025

Actual Word Count: 1000

Extension: N * **Extension Due Date:**

I wish my assignment to be considered for including as an exemplar in the **School Bank of Assessed Work**.

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It is a requirement to complete the first section, and to include this completed template as the first page of every assignment that is submitted for marking (your School will advise on exceptions).

Declaration

By submitting this work, you confirm that it is your own original work, that it reflects your own thinking, analysis, and understanding, and that it therefore complies with the University's Code of Practice on Academic Integrity.

This includes adhering to university guidance on the [ethical use of generative AI](#).

Please tick the box below to confirm this.

☒ I confirm that this is my own original work. I have not included any content from generative AI tools in this assignment, except in ways explicitly permitted in the assignment instructions.

CONTINUED BELOW

Remember to use feedback from previous formative and summative assignments towards this assignment. You can discuss previous feedback with your module or seminar convenors during office hours before submitting. If you wish to receive targeted feedback that will support your learning, please use this second section.

Section One: Reflecting on the feedback that I have received on previous assessments, the following issues/topics have been identified as areas for improvement: (add 3 bullet points). *NB – for first year students/PGTs in the first term, this refers to assessments in your previous institution*

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Section Two: In this assignment, I have attempted to act on previous feedback in the following ways (3 bullet points)

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Section Three: Feedback on the following aspects of this assignment (i.e. content/style/approach) would be particularly helpful to me: (3 bullet points)

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AI-Enhanced Routing for Equitable Food Distribution: A Vietnam Case Study

Current word count (excluding title, bibliography, and appendix): 1000

1 Introduction

Food insecurity in low- and middle-income countries often stems from coordination failures rather than insufficient production: food spoils in surplus areas while nearby regions face shortages (FAO et al., 2025). Public food assistance systems commonly rely on manual, ad-hoc logistics with limited optimisation support (De Vries et al., 2020; Rodríguez-Espíndola et al., 2023). “Last-mile” delivery is a major constraint, particularly in infrastructure-poor areas, generating delays and inconsistent access for the most vulnerable households (Roubert et al., 2018).

Although food security nets aim to allocate resources according to need, state-led systems are frequently criticised for inefficiency (Willis, 2011, Ch. 2). Meanwhile, the private sector has transformed logistics through AI-enabled forecasting, inventory management, and routing, achieving high efficiency (Li, 2024). Recent research in humanitarian logistics shows that reinforcement learning can outperform traditional heuristics in sequential, uncertain allocation settings similar to emergency food distribution (Yu et al., 2021; Van Steenbergen et al., 2023; Yan et al., 2022). However, such tools remain underutilised in real-world development settings, suggesting potential gains from integrating AI into national food distribution systems.

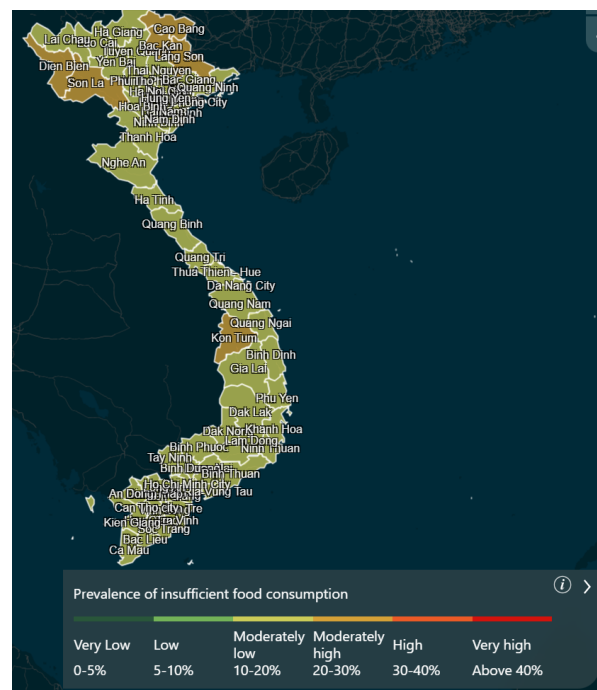
Vietnam illustrates this gap clearly. Despite major poverty reduction, remote ethnic-minority regions remain hard to reach due to structural barriers to markets and distribution (Ngo Chi et al., 2019). Emergency food delivery is still manual and lacks any AI-based prioritisation. The November 2025 floods, which cut off entire districts and left thousands reliant on emergency supplies (AHA Centre, 2025), highlighted the need for more responsive systems. These conditions make Vietnam an appropriate testbed for AI-enhanced routing.

This project compares a traditional cost-minimising routing model with a reinforcement-learning model that prioritises hunger-affected provinces. This project evaluates whether AI-driven routing can improve the equity and efficiency of emergency food distribution in Vietnam, particularly under conditions of isolation, road disruption, and surge demand.

2 Data

Vietnam maintains several centralised food storage nodes, including national food reserve warehouses and large rice facilities in the northern and southern regions. For this simulation, only two depots are modelled: one near Hanoi and one near Cần Thơ/HCMC, representing the country's national food reserve clusters. This simplified structure approximates national distribution patterns and captures the geographic imbalance between where food is stored and where vulnerable populations are concentrated.

Demand is modelled at the provincial level using WFP HungerMap data (World Food Programme, n.d.), which identifies provinces such as Kon Tum, Sơn La, Điện Biên, and Bắc Kạn as having higher insufficient food consumption. Province-level modelling is appropriate because national distribution decisions occur at this scale and sub-provincial data are limited. Each province is assigned a population estimate, hunger severity score, food requirement, and travel cost from each depot.



The routing network is constructed using OpenStreetMap data (OpenStreetMap contributors, n.d.), where intersections serve as nodes and road segments as edges weighted by distance or travel time. This provides a simplified but representative transport network for modelling routes. Terrain or weather attributes may be incorporated in future work.

3 Methodology

Two routing approaches are compared: (1) a baseline cost-minimising heuristic commonly used in humanitarian logistics, and (2) a reinforcement learning (RL) model that prioritises high-need provinces. The RL agent receives state information on location, unmet demand, and hunger severity, and learns equitable routing policies through repeated simulated episodes. Technical justification for these methods are provided in Appendix 1A-1B.

Performance is assessed by distance travelled, equity of deliveries (the proportion of food reaching high-hunger provinces and reductions in unmet demand), unmet need (aggregate and per-province shortfalls), and robustness to disruptions such as road closures or supply shocks. Together, these metrics capture the trade-offs between logistical efficiency and distributive fairness while testing the model's reliability under realistic constraints.

4 Expected Contribution

This study contributes to humanitarian logistics research, where routing systems typically rely on simple heuristics that prioritise speed and distance but cannot incorporate welfare or fairness objectives (Clarke & Wright, 1964; Laporte, 1992; Anaya-Arenas et al., 2014). By demonstrating how reinforcement learning can embed hunger-based prioritisation within a national food distribution system, the project addresses recognised gaps in equity-oriented and dynamic decision-making models (Holguín-Veras et al., 2012). Comparing an equity-aware RL agent with a cost-minimising heuristic provides empirical insight into efficiency–fairness trade-offs rarely quantified in existing work.

Within the RL literature, the study contributes to emerging RL-for-allocation applications by incorporating an explicit fairness objective; most logistics-focused RL models optimise efficiency alone and overlook welfare considerations (Yan et al., 2022).

For policy, the Vietnam case illustrates how national food reserves could be allocated more equitably to remote, high-need provinces, particularly during disruptions. The framework is generalisable and adaptable for other low- and middle-income countries seeking to modernise food distribution systems.

5 Limitations and Ethical Considerations

Several limitations arise from the simplified structure of the simulation. Modelling at the provincial level masks substantial within-province inequality, particularly in mountainous

or ethnically diverse regions, and cannot capture the fine-grained spatial demand variation observed in real disasters. This helps at the provincial level but does not address the “last-mile” delivery gap noted by Roubert et al. (2018).

The model also abstracts from political, institutional, and behavioural constraints that significantly influence allocation outcomes, such as local governance capacity, administrative prioritisation, or budget limitations highlighted in humanitarian logistics literature (Holguín-Veras et al., 2012). Simplifying assumptions regarding transport costs, vehicle capacity, and road accessibility may understate the operational frictions faced during severe weather events or infrastructure disruption. This simplification can worsen broader critiques of RL systems, including instability, limited interpretability, and difficulties in encoding fairness constraints (Yan et al., 2022).

Ethical considerations further complicate the deployment of AI-based allocation models. Prioritisation systems risk reinforcing existing biases if they rely on incomplete or inaccurate need assessments, potentially disadvantaging marginalised populations. Algorithmic allocation raises deeper governance concerns regarding autonomy and the centralisation of decision-making authority (Pałka, 2020). Implementing such systems in public food distribution requires transparency, accountability mechanisms, and human oversight to ensure that results align with social welfare objectives. These constraints underscore that while simulation can illustrate potential benefits, translating RL-based allocation into policy demands careful ethical evaluation and coordination with public institutions.

Reference

Peer-reviewed Articles

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Data sources

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News

<https://www.aboutamazon.eu/news/innovation/ai-in-logistics-the-silent-revolution-transforming-work>

Appendix 1: Technical description of the baseline heuristic model vs Reinforcement Learning's contribution

1A. Heuristic model

The baseline model represents conventional cost-minimising logistics, implemented through a greedy nearest-neighbour heuristic rather than a fully optimised vehicle routing problem (VRP), reflecting the fast, simple decision rules commonly used in practice. Foundational work in vehicle routing established the dominance of such heuristics: Clarke and Wright's (1964) Savings Algorithm demonstrated that iteratively merging delivery points based on mileage savings yields efficient routes with minimal computational burden, setting the template for practical routing methods. Laporte's (1992) survey similarly shows that real-world VRP implementations rely heavily on procedures such as nearest-neighbour, insertion heuristics, sweep algorithms, and Clarke–Wright variants, because exact optimisation is computationally prohibitive for large or time-constrained problems.

In humanitarian logistics, this reliance on heuristics is even more pronounced. Post-disaster distribution operates under rapidly changing demand, damaged or inaccessible transport networks, fragmented decision-making, and the need to minimise deprivation costs rather than purely financial ones (Holguín-Veras et al., 2012). Under such conditions, static optimisation models are often infeasible, and agencies instead adopt simple, fast heuristics that can function with limited information. A systematic review of 83 studies confirms that humanitarian routing overwhelmingly uses heuristic approaches, including greedy, sweep, and other VRP-derived methods, while equity considerations appear rarely and most models assume static, deterministic demand (Anaya-Arenas et al., 2014). These limitations suggest that conventional heuristics, though operationally realistic, cannot represent equity goals or adapt dynamically to changing conditions, motivating the exploration of reinforcement learning as a more responsive and fairness-oriented routing method.

1B. Reinforcement Learning

These limitations of heuristic approaches, namely their static structure, inability to incorporate equity objectives, and lack of responsiveness to changing conditions, motivate the use of reinforcement learning (RL) as a more adaptive routing method. The RL model developed here is designed explicitly to embed equity into the routing process while responding dynamically to unmet demand. The agent observes the current truck location, remaining load, province-level unmet need, and the hunger profile of each

destination, and selects the next delivery point based on a reward function that assigns higher value to serving high-need areas while penalising long or inefficient travel. Deliveries reduce unmet demand within each episode, making the routing problem inherently sequential rather than static.

The agent is trained over repeated simulated episodes until its behaviour stabilises, allowing it to learn policies that balance efficiency and equity under varying conditions. This approach aligns with emerging evidence that RL-based allocation strategies adapt more effectively to uncertainty and dynamic demand than fixed heuristics, and can outperform conventional models in humanitarian or resource-limited contexts (Yu et al., 2021; Van Steenbergen et al., 2023).

Appendix 2: Acknowledgement of AI use

ChatGPT was used heavily during the process of drafting this research proposal. Specifically, the tool was asked to help with all the steps a human research assistant can do, including:

Task in Workflow	Prompt Given	Use and Modification of Output
Co-brainstorming to formalize initial research idea	“I want to research AI for food allocation. How to turn it into a testable research?”	Output was used to clarify the broad research direction.
Identify actual, real-world challenges with the global and Vietnam food system	“... Take a read at my [previous] research proposal for the assignment. [...] I usually [...] try to solve a non-existing problem. What do you think? I'm thinking of identifying an ACTUAL problem, Vietnam or elsewhere” + Word file	Verified against external literature and rewrote them into Paragraph 4 of the Introduction.
Looking up relevant material	“can you name the reports that specifically say this? Many low- and middle-income countries lack a formal, algorithmically optimized food distribution system. [etc]”	Selected FAO et al. (2025) from many sources. Similar prompts were used to locate sources for heuristic routing (Appendix 1A).
Proof-reading and language tightening	Example: “Food insecurity remains one of the most pressing challenges in global development, [...] raw paragraphs]. is this good”	Similar prompts are used throughout the proposal for stylistic tightening.
Summarising articles	“summarize this pls does it support this claim: [claim]” + PDF file of article	Cross-checked for factuality, and then rewritten.

Citation formatting	“help me cite that report in APA” + PDF of report	Bibliography
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